To run **YOLOv8** applications, you will need to install the required libraries in an environment like `**yolov8**`. The step-by-step process you need to follow to run **YOLOv8** is explained below.

Step-by-Step YOLOv8 Installation and Execution:

1. Create a New Environment in Anaconda Prompt (If You Haven't Already):

If you haven't created an environment before, start by creating one named **yolov8.**
(You can follow the steps mentioned above.)

```
conda create --name yolov8 python=3.10
conda activate yolov8
```

2. CUDA & PyTorch Installation::

YOLOv8 requires CUDA to utilize GPU acceleration. If you have a CUDA-compatible GPU, you can install the CUDA-supported version of PyTorch using the command below.
If you're not using a GPU, you can also install the version without CUDA support.

Installing PyTorch with CUDA support:

```
conda install pytorch torchvision torchaudio cudatoolkit=11.3 -c pytorch
```

If you don't want to use a GPU, you can install the CPU-only version of PyTorch with the following command:

```
conda install pytorch torchvision torchaudio cpuonly -c
pytorch
```

3. Installing the YOLOv8 Library:

The required library for YOLOv8 is **ultralytics**. To install YOLOv8, you can use the following command:

```
pip install ultralytics
```

4. Testing with the YOLOv8 Application:

After installing YOLOv8, you can perform a quick test using the commands below.

Enter the Python environment:

```
python
```

Run a simple test using the YOLOv8 model:

```
from ultralytics import YOLO

# Load a pretrained YOLOv8 model
model = YOLO('yolov8n.pt')  # 'yolov8n.pt' is a small and fast model. Other models can
also be used.
```

```
# Load a test image and make predictions
results = model('path_to_image.jpg')  # Replace with the path to your test image
results.show()  # Display the image with predictions
```

- Here, yolov8n.pt is a **lightweight version of YOLOv8.** You can also try other versions like yolov8s.pt, yolov8m.pt, etc., depending on your needs.
- Just replace 'path_to_image.jpg' with the actual path to your image file to test

5. Installing Additional Required Libraries:

In addition to the main libraries, YOLOv8 may also require the following:

```
pip install opencv-python matplotlib
```

- **OpenCV**, Used for processing and displaying images
- **Matplotlib**, Used for visualizing images and results

6. Training the Model (Optional)

If you want to train the model using your own dataset, you can use the command below.
However, you'll first need to prepare your dataset in YOLO format.

Train the model using:

```
model.train(data='path_to_your_dataset.yaml', epochs=10)
```

In this command:

- `'path_to_your_dataset.yaml'` refers to the **YAML** file containing paths to your images and their labels.
- `epochs=10` means the model will be trained for 10 epochs.

7. Using YOLOv8 Outputs

Whether you use a pretrained or a custom-trained model, you can use its outputs to make predictions and visualize them.
The output will include detected objects with their labels and bounding box coordinates.

Installing All Required Libraries from a `requirements.txt` file

1. Go to the directory containing the `requirements.txt`:

- Use the terminal or Anaconda Prompt to navigate to that directory.

2. Run the following command to install all listed libraries:

You can install all the required libraries listed in a `requirements.txt` file using the following command

```
pip install -r requirements.txt
```

This command will install all libraries and the specified versions mentioned in the `requirements.txt` file

`requirements.txt` File Format:

A typical `requirements.txt` file lists libraries in the following format:

```
numpy==1.21.0
pandas>=1.3.0
scikit-learn==0.24.2
torch==1.9.0
```

- **==**: Installs the exact version
- **>=**: Installs the specified version and above
- **<=**: Installs the specified version and below

Summart:

1. Go to the directory where the `requirements.txt` file is located.
2. `pip install -r requirements.txt` Run the mentioned command to install all libraries.

This command will install all the necessary libraries into your environment.

Summary:

1. **Create and activate a new Anaconda environment.**
2. **Install PyTorch and CUDA (if you want GPU support).**
3. **Install YOLOv8 using `pip install ultralytics` command.**
4. **Load and test the YOLOv8 model in the Python environment.**
5. **Install additional libraries like OpenCV and Matplotlib.**
6. **If you want to train the model with your own data, prepare your dataset in YOLO format and use the training command.**

By following these steps, you can successfully set up, run, and test **YOLOv8** applications on your computer.

To create a new environment in Anaconda Prompt and install the EasyOCR library, you can follow the steps below. Python 3.10 is compatible with EasyOCR, but for better compatibility, Python 3.8 or 3.9 is also commonly preferred. In this guide, we'll proceed with Python 3.10.

Step-by-Step: Creating an Anaconda Environment for EasyOCR

1. Launch Anaconda Prompt:

- Launch Anaconda Prompt. If anaconda is not installed you can download and install it from the official website: https://www.anaconda.com/products/distribution

2. Create a new Environment named ocr with python 3.10:

Using the following command create a new environment named ocr with python=3.10

```
conda create --name ocr python=3.10
```

- This command will create an environment named ocr with python=3.10 version
- If you prefer a more compatible version (like Python 3.8 or 3.9), you can replace `python=3.10` with `python=3.8` or `python=3.9` .

3. Activate the Environment:

To activate the created environment

```
conda activate ocr
```

4. Install EasyOCR Library:

Once the environment is active, install EasyOCR using, `pip` like:

```
pip install easyocr
```

5. Test EasyOCR Installation:

You can test whether EasyOCR is working correctly by running the following Python code:

```
python
```

Once Python shell opens, test EasyOCR by running the following command

```
import easyocr

reader  = easyocr.Reader(['en'])     # 'en'  English Language Support
result = reader.readtext('path_to_image.jpg')   # Provide an image path to test

print(result)
```

This will read and print the text from an image.

6. Install Other Required Libraries

EasyOCR may require some additional dependencies to function properly.
If you encounter any missing libraries or errors, you can install them using the following commands:

```
pip install torch torchvision
pip install matplotlib
```

Extra Notes:

- If you want to use a specific **OCR model** or language support (e.g. Turkish), you can specify language code like `[tr]` :
- `reader = easyocr.Reader(['en', 'tr'])`

By following these steps, you can create an environment named ocr and start using EasyOCR successfully.
